

Merkel cell carcinoma: value of sentinel lymph-node status and adjuvant radiation therapy

A. Servy^{1,†}, E. Maubec^{2,3,4*,†}, P. E. Sugier^{4,5,6}, F. Grange⁷, S. Mansard⁸, T. Lesimple⁹, E. Marinho¹⁰, B. Couturaud¹¹, A. Girod¹¹, S. Albert^{6,12}, R. Dendale¹³, E. Calitchi¹⁴, L. Sarda^{6,15}, J. Chanal¹, L. Deschamps¹⁰, X. Sastre-Garau¹⁶, L. Laroche^{2,3}, B. Crickx^{6,17} & M. F. Avril^{1,18}

¹Department of Dermatology, APHP, Hôpital Cochin, Paris; ²University of Paris 13, Bobigny; ³Department of Dermatology, APHP, Hôpital Avicenne, Bobigny; ⁴UMR-946 INSERM, Genetic Variation and Human Diseases Unit (UMR-946), Paris; ⁵University of Pierre et Marie Curie, Paris; ⁶University of Paris Diderot, Paris; ⁷Department of Dermatology, Hôpital Robert-Debré, Reims; ⁸Department of Dermatology, CHU Estaing, Clermont-Ferrand; ⁹Department of Oncology, Centre Eugène-Marquis, Rennes; ¹⁰Department of Pathology, APHP, Hôpital Bichat, Paris; ¹¹Department of Surgery, Institut Curie, Paris; ¹²Department of Otorhinolaryngology, APHP, Hôpital Bichat, Paris; ¹³Department of Oncology, Institut Curie, Paris; ¹⁴Department of Radiotherapy, Clinique de la Porte de Saint-Cloud, Boulogne; ¹⁵Department of Nuclear Medicine, APHP, Hôpital Lariboisière, Paris, Descartes; ¹⁶Institut Curie, Department of Pathology, Paris; ¹⁷Department of Dermatology, APHP, Hôpital Bichat, Paris; ¹⁸University of Paris Descartes, Paris, France

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Background: Sentinel lymph-node (LN) biopsy (SLNB) is a valuable tool to assess the regional LN status in Merkel cell carcinoma (MCC). However, its prognostic value is still debated. This study was undertaken to assess SLNB usefulness for MCC management and to determine the impact of SLNB status on disease-free survival (DFS) and overall survival (OS) by comparing SLNB-positive versus -negative patients according to demographic, clinical and treatment characteristics.

Patients and methods: In this retrospective, multicenter observational study, SLNB was proposed to all patients referred for clinically N0 MCC. Treatment schedule consisted of wide-margin surgical resection of primary MCC followed by adjuvant radiation therapy (aRT) to the primary site and, for SLNB-positive patients, radical LN dissection followed by regional aRT. Univariate and multivariate analyses determined factors associated with DFS and OS.

Results: Among 87 patients with successful SLNB, 21 (24.1%) were SLNB-positive. Median follow-up for the entire series was 39 months; respective 3-year DFS and OS rates were 73% and 81.4%, respectively. Univariate analysis (all patients) identified SLNB-negativity as being associated with prolonged OS ($P=0.013$) and aRT (all sites considered) was associated with longer DFS ($P=0.004$) and OS ($P=0.018$). Multivariate analysis (all patients) retained SLNB status and aRT (all sites considered) as being associated with improved DFS ($P=0.014$ and 0.0008) and OS ($P=0.0020$ and 0.0019). Moreover, for SLNB-negative patients, tumor-bed irradiation was also significantly associated with prolonged DFS ($P=0.006$) and OS ($P=0.014$).

Conclusion(s): The present study demonstrates that SLNB-negativity is a strong predictor of longer DFS and OS in stage I and II MCC patients. The similar benefit for aRT on tumor bed observed in this study has to be confirmed by a prospective study. The results advocate for SLNB being considered to all MCC patients.

Key words: Merkel cell carcinoma, adjuvant radiation therapy, sentinel lymph-node biopsy, overall survival, disease-free survival

introduction

Merkel cell carcinoma (MCC) is a rare neuroendocrine skin cancer that develops mainly in elderly and immunocompromised individuals. The association of Merkel cell polyomavirus

with 80% of MCC suggests a viral etiology [1]. MCC incidence has risen over the last few decades worldwide, with ~1500 new cases diagnosed every year in the USA [2, 3]. MCC may have an aggressive course, with 10-year survival rates ranging between 50% and 65% [4]. Prognostic factors are female sex, anatomic site, tumor size and, most importantly, American Joint Committee on Cancer (AJCC) stage. Ten-year survival rates are estimated at 71% for patients with localized disease and 49% for those with regional lymph-node (LN) metastases [4].

*Correspondence to: Prof. Eve Maubec, Department of Dermatology, Hôpital Avicenne, 125, rue de Stalingrad, 93009 Bobigny, France. Tel: +33-1-48-95-77-07; Fax: +33-1-48-95-51-87; E-mail: eve.maubec@inserm.fr

[†]These authors contributed equally to the work.

MCC metastases develop first in regional LNs and are observed in about one-third of the patients at initial staging [4]. Sentinel LN biopsy (SLNB) has been shown to be a valuable tool to assess the regional LN status in MCC patients without palpable LN. SLNB detects metastases in 20%–30% of these patients [5, 6]. Distinction between macroscopic and microscopic regional LN involvement has been incorporated into the most recent AJCC staging of MCC [7]. SLNB-positivity is associated with several histologic characteristics, including lymphovascular invasion, tumor thickness, mitotic rate and histologic growth pattern [8, 9].

However, the value of SLNB is still debated [10]. Studies on the impact of SLNB on relapse and survival are rare, and results are discrepant [5, 6, 8, 11–14]. Moreover, optimal MCC management strategies have not yet been clearly defined. Although a local excision of the primary tumor with wide margins is usually recommended, no consensus has been reached concerning adjuvant radiation therapy (aRT) [15].

The aim of this study was to determine SLNB usefulness for the management of MCC patients and, more precisely, to determine the impact of SLNB status on disease-free survival (DFS) and overall survival (OS), by comparing a large series of SLNB-positive and -negative patients according to their demographic, clinical and treatment characteristics.

patients and methods

This multicenter, retrospective, observational study was conducted in six French centers. It was approved by the Île-de-France III institutional review board (no. SC 3199). SLNB was offered to all consecutive patients referred for histologically confirmed MCC without palpable regional LNs or metastatic disease (cN0M0) diagnosed between November 1997 and December 2013, including seven previously reported patients [16]. Demographic, clinical and histological data of all patients who underwent SLNB were computerized, after obtaining informed consent. Patients were classified into two groups according to positive (SLNB+) or negative (SLNB–) SLNB status.

treatment and follow-up

The initial work-up included physical examination and at least a chest X-ray and abdominal and pelvic ultrasonography, and/or a computed tomography (CT) scan and/or PET–CT to confirm the cN0M0 stage. For all patients, the treatment schedule included resection of the primary tumor with a wide macroscopic margin (2–3 cm) and SLNB. After technetium-99 injections close to the primary MCC biopsy resection scar or residual tumor, preoperative lymphoscintigraphy identified the SLN [16]. Histologic evaluation included serial sectioning, hematoxylin and eosin staining, and cytokeratin-20 and pancytokeratin immunolabeling. Histologic examination confirmed the complete primary MCC excision for all patients. After healing of surgical wounds, the therapeutic protocol included aRT delivery of 50 Gy to the primary MCC site. SLNB– patients received no regional aRT. For SLNB+ patients, regional treatment, according to the protocol, consisted of radical LN dissection followed by regional delivery of aRT (50 Gy). No systemic adjuvant chemotherapy was given. Thereafter, follow-up visits were scheduled every 3 months and consisted of physical examinations and, depending on any new symptoms or for stage III patients, radiologic investigations.

data collection

Patient demographics, tumor characteristics, treatments and outcomes were collected retrospectively from medical records. Demographic and clinical features included sex, age at diagnosis, time period of SLNB, primary tumor site, preoperative work-up and immunocompromised status (hematologic malignancies or immunosuppressive therapy). Tumor size (largest dermatologist-measured diameter in centimeters), macroscopic-free margins at surgical resection, pathological-free margins, SLNB status, complementary radical LN-dissection status (positive versus negative) for SLNB+ patients, aRT delivery to the primary MCC site and/or to the regional LN basin were also recorded. MCC stage was assigned according to the 2009 AJCC classification [7]. In addition, follow-up information including first recurrence type (local, in-transit, regional and distant) and its date, defined as the first confirmation of recurrence, and death (specific or not) were retrieved from medical records. Follow-up data were recorded until 15 December 2014.

statistical analyses

Descriptive data are expressed as means (ranges) or median (IQR) for continuous variables or number (%) for categorical data. Chi-square tests were used to compare qualitative variables and Student's *t*-tests for quantitative variables. Univariate DFS and OS were calculated with a Cox regression model, with χ^2 assessment of significance. Variables statistically significant ($P < 0.1$) in univariate analysis were entered into a Cox regression analysis with a backward-selection method to determine independent associations with DFS or OS. All analyses were computed with SAS software version 9.3.

results

Ninety-one consecutive patients with clinically N0 MCC underwent SLNB. The procedure failed in four patients with primary head and neck tumors (3 surgical failures and 1 lymphoscintigraphy failure) who were excluded from these analyses.

SLNB status and patient characteristics

Among 87 patients [38 (43.7%) men; mean age at MCC diagnosis, 70.7 (range, 31–90) years] with successful SLNBs (Table 1), 21 had metastatic SLNs (24.1%). Fifty-seven percent of MCCs were located on extremities. Nine percent of the patients were immunocompromised in some way. Most of the patients (60.9%) had T1 MCC <2 cm in diameter. Initial work-up consisted of at least a thoracic and abdominal CT scan in 67.4% of patients, at least a PET–CT scan in 40.7% of patients and both CT and PET–CT scan in 24% of patients. Sixteen patients (18.6%), with a median age of 79 years, had only a chest X-ray and an abdominal ultrasound at initial staging. Ninety-two percent of patients had re-excision margins ≥ 2 cm. Excision margin was pathologically free in all cases. Three cases achieved only narrow margins, but there were none R+ excision.

The median number of excised SLN was 2 (1–9) and the median number of positive SLN was 1 (1–4). The frequency of men was significantly higher in SLNB+ (66.7%) than in SLNB– (36.4%, $P = 0.01$). The median follow-up was longer in SLNB– patients (41.3; IQR, 29–66) than in SLNB+ patients (27.1; IQR, 19–51; $P = 0.026$). Other characteristics were comparable for the two groups. The frequencies of SLNB+ patients did not differ between T1 and $\geq T2$ stages ($P = 0.4$) or among T1 stages between MCC <1 cm in diameter versus 1–2 cm ($P = 0.5$). Notably, 6 (24%) of the 25 patients with MCC diameters <1 cm were SLNB+.

Table 1. Baseline characteristics of 87 MCC patients with successful SLNB procedures

Characteristic	All patients (N = 87)		SLNB+ (N = 21)		SLNB- (N = 66)		P-value
	No. of patients	%	No. of patients	%	No. of patients	%	
Males	38	43.7	14	66.7	24	36.4	0.014
Age							
Mean (years)	70.7		71.0		70.6		0.90
Range	31–90		37–89		31–90		
≥70 years	54	62.1	12	57.1	42	63.6	0.39
Primary MCC location							
Head and neck	27	31.0	5	23.8	22	33.3	0.39
Extremities	50	57.5	13	61.9	37	56.1	
Trunk	10	11.5	3	14.3	7	10.6	
T stage ^a							
T1 ≤1 cm	25	28.7	6	28.6	19	28.8	0.43
T1 >1 cm	28	32.2	6	28.6	22	33.3	
≥T2	34	39.1	9	42.9	25	37.9	
Preoperative work-up							
FDG-PET/CT ^b	35	40.7	8	38.1	27	41.5	1
Abdominal and thoracic CT scans ^b	58	67.4	14	66.7	44	67.7	1
Chest X-ray and abdominal US only	16	18.6	3	14.3	13	20	0.75
NA	1		0		1		
Immunocompromised							
Yes	8	9.2	3	14.3	5	7.6	0.29
SLNB time period							
1999–2002	7	8.1	3	14.3	4	6.1	0.35
2003–2006	17	19.5	5	23.8	12	18.2	
2007–2010	37	42.5	6	28.6	31	47.0	
2011–2014	26	29.9	7	33.3	19	28.8	
Follow-up							
Median (months)	39.3		27.1		41.3		0.026
IQR	22–64		19–51		29–66		

^aAccording to American Joint Commission on Cancer, 7th edition [7].^bAt least one FDG-PET/CT or at least abdominal and thoracic CT scans.

MCC, Merkel cell carcinoma; SLNB, sentinel lymph-node biopsy; NA, not available; IQR, interquartile range; US, ultrasound; IQR, interquartile range.

additional treatments according to SLNB status

Treatment according to SLNB status is shown in Figure 1. Twenty (95%) of the 21 SLNB+ patients underwent radical LN dissection, which led to the identification of additional metastatic LNs in 6/20 (30%). The median number of removed LN in additional LN dissection was 11 (0–33). All together, the median number of resected regional LN was 13 (1–37) and the median number of positive LN was 1 (1–8). Radical LN dissection was followed by aRT delivery to the primary site and involved LN basin for 18 patients, only to primary tumor site in one patient. aRT could not be delivered in one patient due to early death. The last SLNB+ patient refused radical LN dissection, but received aRT to both primary and regional sites. Fifty-five (83%) SLNB- patients received aRT to the primary MCC site. Eleven SLNB- patients did not receive aRT on the primary tumor bed for the following reasons: early relapse in two patients, medical decisions in seven patients and aRT refusal in two patients. Overall, aRT was delivered to the tumor bed of 95% of SLNB+ patients and 83% of those SLNB-. Moreover, aRT was delivered to regional sites of 90% of SLNB+ patients. The baseline characteristics of patients who received and who

have not received aRT are shown in supplementary Table S1, available at *Annals of Oncology* online. Characteristics were comparable for the two groups, except SLNB period of time.

outcomes

Follow-up data are presented in Figures 1 and 2. Within a median of 12 (range, 2–43) months, 24.1% of all patients developed recurrences. Three first recurrences were local, 3 were in-transit metastases, 5 were regional LN metastases (i.e. presumed false-negative SLNB findings) and 10 were distant metastases.

The 3-year DFS rate was 73%. Univariate analysis of the entire series identified aRT (all sites considered, $P = 0.01$), but not SLNB status ($P = 0.08$), as being associated with prolonged DFS (Table 2). For SLNB- patients (Table 2), aRT to the primary MCC site was also significantly associated with longer DFS ($P = 0.004$). Multivariate analyses (Table 3) retained SLNB status and aRT delivery (all sites considered) as being significantly associated with improved DFS (respectively, $P \leq 0.014$ and 0.0008).

The 3-year OS rate was 81.4%: 88% for SLNB- patients versus only 57.2% for SLNB+ patients. Sixteen patients died of MCC and

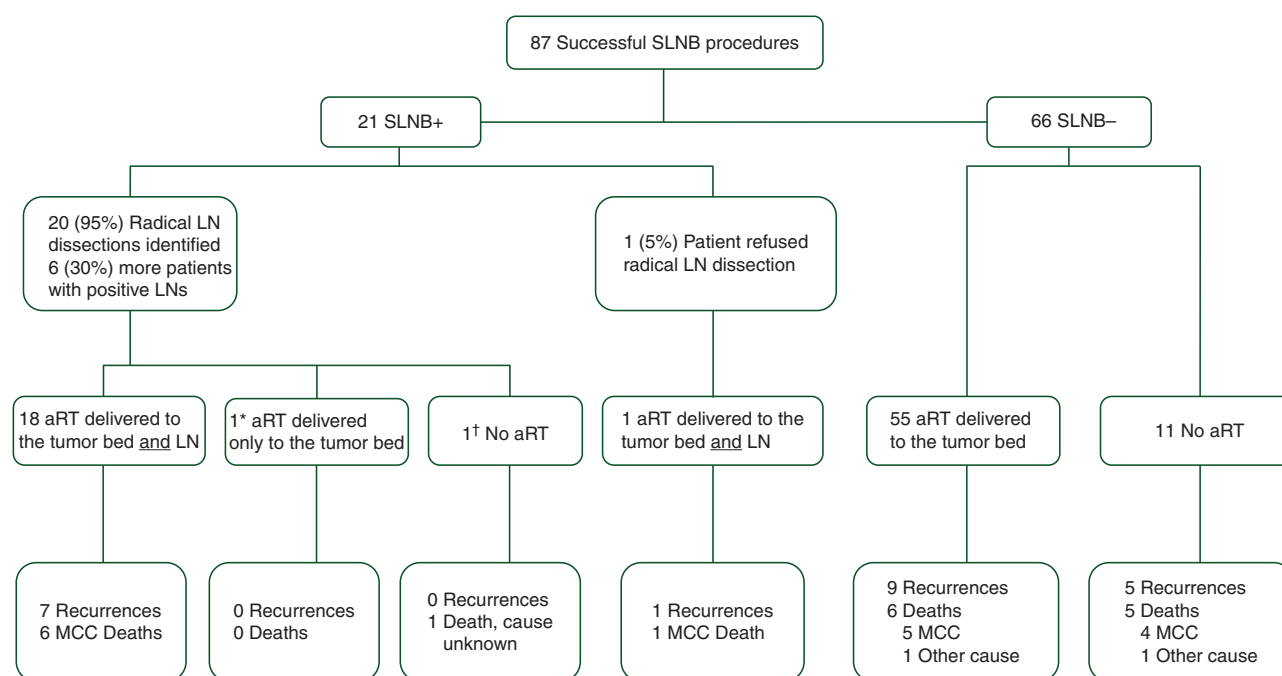


Figure 1. Merkel cell carcinoma treatments and outcomes according to sentinel lymph-node biopsy (SLNB) status. aRT, adjuvant radiation therapy. Asterisk indicates that one SLNB+ patient had delayed healing of the regional surgical wound precluding aRT delivery. Dagger indicates that one SLNB+ patient died of an unknown cause before aRT.

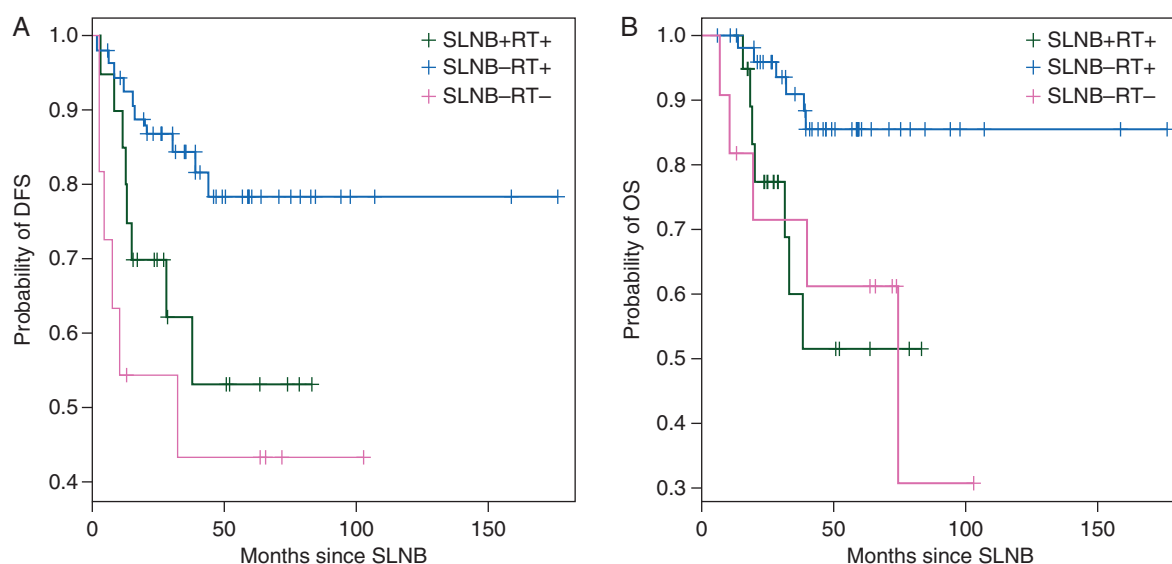


Figure 2. (A) Disease-free survival (DFS) and (B) overall survival (OS) according to sentinel lymph-node biopsy (SLNB) status and adjuvant radiation therapy (aRT) delivery. SLNB- patients who received aRT (SLNB-aRT+) had better DFS and OS than SLNB-aRT- and SLNB+aRT+ patients. The only SLNB+aRT- patient is not represented.

three of other or unknown causes. The median times from initial treatment to death or specific death were, respectively, 20.5 (range, 2–74) and 24.4 (range, 6.9–24.4) months. Univariate analysis identified female sex ($P = 0.027$), SNLB-negativity ($P = 0.013$) and aRT ($P = 0.018$) as being associated with longer OS (Table 2). aRT delivery to the tumor bed of SLNB- patients was also associated with prolonged OS ($P = 0.014$). Multivariate analyses of all patients retained SLNB status ($P = 0.0020$) and aRT (all sites considered, $P = 0.0019$) as being significantly associated with OS (Table 3).

discussion

To the best of our knowledge, this study on a large homogeneous series of 87 MCC patients who underwent SLNB is the first to explore DFS and OS according to SLNB status, demographic and clinical characteristics and delivery of aRT. SLNB- status was retained as a strong predictor of longer DFS and OS for patients with stage I or II MCCs. In addition, our results also suggest that aRT might also be associated with longer DFS and OS in these

Table 2. Univariate hazard ratios for disease-free survival and overall survival of all patients with SLNB or the SLNB–

Characteristic	All patients (N = 87)				SLNB– (N = 66)			
	Disease-free survival hazard ratio (95% CI)	P-value	Overall survival hazard ratio (95% CI)	P-value	Disease-free survival hazard ratio (95% CI)	P-value	Overall survival hazard ratio (95% CI)	P-value
Age at diagnosis	1.00 (0.99–1.01)	0.26	1.00 (1.00–1.01)	0.11	1.00 (1.00–1.01)	0.10	1.01 (1.00–1.01)	0.10
Sex (male versus female)	0.49 (0.22–1.08)	0.077	0.33 (0.13–0.89)	0.027	0.55 (0.21–1.48)	0.24	0.30 (0.09–1.03)	0.06
SLNB time period	0.94 (0.60–1.47)	0.78	0.85 (0.51–1.43)	0.55	0.94 (0.53–1.68)	0.84	0.90 (0.44–1.85)	0.77
Adjuvant radiation therapy ^a (yes versus no)	0.28 (0.12–0.67)	0.004	0.31 (0.12–0.82)	0.018	0.24 (0.09–0.67)	0.006	0.22 (0.07–0.74)	0.014
Chest X-ray and abdominal US only	1.23 (0.46–3.28)	0.69	1.62 (0.53–4.94)	0.40	2.14 (0.74–6.18)	0.16	3.25 (0.94–11.30)	0.064
Immunocompromised (yes versus no)	1.18 (0.27–5.13)	0.82	1.18 (0.27–5.13)	0.82	2.23 (0.51–9.86)	0.29	1.38 (0.18–10.85)	0.76
SLNB+ versus SLNB–	2.08 (0.92–4.73)	0.079	3.20 (1.27–8.01)	0.013	–	–	–	–
Stage (T1 versus T2 or T3)	1.11 (0.50–2.47)	0.80	1.30 (0.52–3.23)	0.57	0.76 (0.26–2.20)	0.62	1.04 (0.30–3.55)	0.95
Location (head and neck versus trunk and limbs)	0.99 (0.37–2.60)	0.98	0.99 (0.37–2.60)	0.98	0.89 (0.31–2.56)	0.82	0.68 (0.18–2.57)	0.57

^aAccording to SLNB status, aRT was delivered to the primary tumor bed alone or to the primary tumor bed and regional lymph nodes. MCC, Merkel cell carcinoma; SLNB, sentinel lymph-node biopsy; aRT, adjuvant radiation therapy; US, ultrasound.

Table 3. Best multivariate model candidates for disease-free survival and overall survival for all 87 patients

Variables	Disease-free survival hazard ratio (95% CI)	P-value	Overall survival hazard ratio (95% CI)	P-value
aRT ^a (yes versus no)	0.19 (0.07–0.50)	0.0008	0.16 (0.05–0.51)	0.0019
SLNB status (+ versus –)	3.15 (1.26–7.84)	0.014	5.68 (1.89–17.05)	0.0020

^aAccording to SLNB status, aRT was delivered to the primary tumor bed alone or also to regional lymph nodes. aRT, adjuvant radiation therapy; SLNB, sentinel lymph-node biopsy.

patients. The present study confirmed that SLN metastases do indeed occur in MCCs <1 cm in diameter (6 of 26 patients, i.e. 28.6%), as previously reported [8, 9]. Three-year OS rates were higher for the whole series (81.4%) and SLNB– patients (88%) than for the SLNB+ group (57.2%). These high survival rates in patients with local disease, as reported by Lemos et al. [17], might reflect the homogeneous treatment of primary MCC, including histologically complete excision followed by local aRT for most patients. Other published series often combined various primary MCC treatment schemes (complete and incomplete surgical excisions, aRT or not), rendering comparison of survival rates difficult. Only three studies that analyzed OS and DFS are available and their results are discordant [8, 13, 14]. The results of our analysis suggest that SLNB status has a prognostic value for OS and DFS, as in the recent meta-analysis by Sadeghi et al. [14].

As in four of the five published studies [5, 6, 8, 13, 14], SLNB-negativity was a strong predictor of longer DFS according to our multivariate analysis. Indeed, one of the two available monocenter studies that included 153 patients with 41 months of follow-up did not find an association between SLNB status and DFS [8]. Definitions of recurrence differed among studies: two studies considered only recurrence, the LN basin or distant sites, whereas others also included local recurrences and in-transit metastases.

The prognostic value of SLNB status for OS is still debated. Four studies are available: the single-center study on 153 patients cited above, two register-based studies that enrolled, respectively, 474 and 33 patients and a meta-analysis of 172 patients [8, 12–14]. The results of the largest register-based study [12] and the meta-analysis [14] demonstrated survival differences according to SLNB status, as we did. Indeed, here, SLNB+ patients had shorter OS than SLNB– patients. SLNB status and aRT were associated with better specific survival ($P = 0.0034$ and 0.0076 , respectively; data not shown), like in the recent study by Kachare et al. [12] for SLNB status. Although death from other causes might be common in this elderly population, 84% of our patients’ deaths were MCC-specific.

MCC is a radiosensitive tumor. Currently, aRT delivery to the tumor bed is proposed by some clinicians because of the high risk of MCC recurrence. It is considered a postsurgical therapeutic option according to most recommendations [18]. The prognostic value of aRT for DFS and OS is still undetermined, even for analyses based on large register databases [15, 19, 20–24]. The only prospective randomized trial to evaluate the prophylactic impact of aRT delivery to the tumor bed and the regional LN basin after resection of localized disease found less frequent regional relapse but no survival benefit [21].

Our results suggest that aRT of the primary MCC site provided a survival advantage for MCC patients without palpable

LN. This benefit was observed for SLNB+ patients, while the regional aRT impact on SLNB+ patients is more difficult to assess due to the small number of treated patients ($n = 19$) and to the delivery of regional nodal irradiation after radical LN dissection in all patients except one.

Current MCC treatment recommendations include wide-margin surgical excision of the primary tumor. SLNB and aRT are both options in the National Comprehensive Cancer Network guidelines [18]. According to our results, delivering aRT to the primary MCC site might be worthwhile. Further prospective randomized studies are needed to assess definitively the prognostic value of aRT for SLNB+ patients.

The main limitations of this study are the retrospective design and the limited number of patients. However, our patients' clinical characteristics are similar to those previously reported, except for more women and primary MCC limb locations that have also been reported in other European studies on MCC [13, 21]. The SLNB+ patients who did not undergo aRT represents 16.6% of the SLNB+ group. However, this percentage reflects the real-life conditions of the study. The high occurrence of events in this group of 11 patients (54% of patients have at least one event) cannot be incidental and suggests an effect of aRT on the outcome.

In conclusion, the present study demonstrates that SLNB+ status is a strong predictor of longer DFS and OS in stage I and II MCC patients. The similar benefit for aRT on primary MCC site observed in this study has to be confirmed by a larger prospective study. The results advocate for SLNB being proposed to all MCC patients, regardless of tumor size.

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disclosure

The authors have declared no conflicts of interest.

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